Finding Inequalities from Regions

Exam code: 0607

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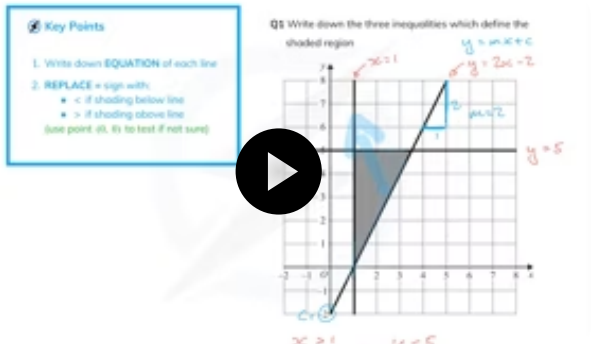

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Help me understand

Interpreting graphical inequalities

How do I know which inequalities are shown on a graph of shaded regions?

- To **identify the inequalities** represented by the shaded regions on a graph:
  - Find the **equation** of each line on the graph
    - You may have to calculate the **gradient** and find the **y-intercept** to use $y = mx + c$
    - **Vertical** lines have the form $x = k$
    - **Horizontal** lines have the form $y = k$
  - **Remember** that lines are drawn using:
    - a **solid** line for ≤ or ≥, indicating a line **included** in the region
    - a **dotted** line for < or >, indicating a line **not included** in the region
  - **Replace** the = sign with the relevant inequality
    - ≤ or < if region is **below** line
    - ≥ or > if region is **above** line
    - (Use a point to test if not sure)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

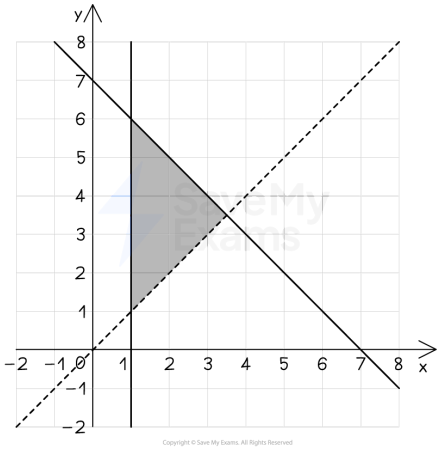
- (Use a point to test if not sure)

Examiner Tips and Tricks

Always read the question carefully to see if the diagram shades the wanted region or the unwanted region.

Worked Example

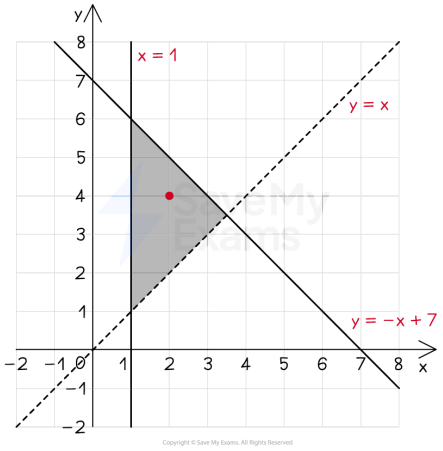
Write down the three inequalities which define the shaded region shown on the axes below.



Answer:

Find the equations of the three lines shown (ignoring inequality signs for now)

You may be able to see the lines $x = 1$ and $y = x$
The other line has the form $y = mx + c$ with y-intercept 7 and gradient -1



dotted, so the inequality is

$$y > x$$

If unsure, check by substituting in coordinates from the shaded region
For example, using (2, 4) as marked on the graph above

"$4 > 2$" is true, so the inequality $y > x$ is correct

For $y = -x + 7$, the shaded region is below the line, and the line is solid, so the inequality is

$$y \leq -x + 7$$

Again, check by substituting (2, 4) into the inequality

"$4 \leq -2 + 7$" is true, so the inequality $y \leq -x + 7$ is correct

For $x = 1$, the shaded region is to the right of the solid line so the inequality is

$$x \geq 1$$

Write all three inequalities together as your final answer

$y > x$, $y \leq -x + 7$ and $x \geq 1$

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